



Trigonometric Form of a Complex Number

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Learning Objectives

- Find the absolute value (modulus) of a complex number
- Convert complex numbers from rectangular form to trigonometric form
- Compare distances of complex numbers from the origin

Show all work and express angles in degrees rounded to the nearest tenth when not exact.

1. Find the absolute value of the complex number.

$$|3 + 2i|$$

Answer: _____

2. Find the absolute value of the complex number.

$$|-2 + 5i|$$

Answer: _____

3. Between $(-2+5i)$ and $(1-6i)$, which one is closer to the origin? Justify with absolute values.

$$|-2 + 5i| \text{ vs. } |1 - 6i|$$

Answer: _____

4. Convert the complex number to trigonometric form (round theta to the nearest tenth of a degree).

$$4 + 3i$$

Answer: _____

5. For $3-5i$, find its absolute value and convert it to trigonometric form (round theta to the nearest tenth).

$$3 - 5i$$

Answer: _____

6. Convert the complex number to trigonometric form.

$$1 + i$$

Answer: _____

7. Convert the complex number to trigonometric form.

$$-2 + 2\sqrt{3}i$$

Answer: _____



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8. Convert the complex number to trigonometric form.

$$-3 - 3i$$

Answer: _____

9. Find the absolute value of the complex number.

$$|6 - 8i|$$

Answer: _____

10. Convert the complex number to trigonometric form (round theta to the nearest tenth).

$$5 - 12i$$

Answer: _____





Remind students that $r = \sqrt{a^2 + b^2}$ and $\tan(\theta) = b/a$, with attention paid to the correct quadrant for θ .

Solutions

1. Find the absolute value of the complex number.

$$|3 + 2i|$$

→ Use the formula $|a+bi| = \sqrt{a^2 + b^2}$.

→ Substitute $a = 3$ and $b = 2$ to get $\sqrt{9 + 4}$.

→ Simplify to $\sqrt{13}$.

Answer: $\sqrt{13}$

2. Find the absolute value of the complex number.

$$|-2 + 5i|$$

→ Apply $|a+bi| = \sqrt{a^2 + b^2}$.

→ Substitute $a = -2$ and $b = 5$ to get $\sqrt{4 + 25}$.

→ Simplify to $\sqrt{29}$.

Answer: $\sqrt{29}$

3. Between $(-2+5i)$ and $(1-6i)$, which one is closer to the origin? Justify with absolute values.

$$|-2 + 5i| \text{ vs. } |1 - 6i|$$

→ Compute $|-2+5i| = \sqrt{4 + 25} = \sqrt{29}$.

→ Compute $|1-6i| = \sqrt{1 + 36} = \sqrt{37}$.

→ Since $\sqrt{29} < \sqrt{37}$, the point $-2+5i$ is closer to the origin.

Answer: $-2 + 5i$ is closer since $\sqrt{29} < \sqrt{37}$

4. Convert the complex number to trigonometric form (round θ to the nearest tenth of a degree).

$$4 + 3i$$

→ Find $r = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$.

→ Find θ with $\tan(\theta) = 3/4$, so $\theta \approx 36.9^\circ$.

→ The point lies in Quadrant I, so θ stays at 36.9° .

→ Write $5(\cos 36.9^\circ + i \sin 36.9^\circ)$.

Answer: $5(\cos 36.9^\circ + i \sin 36.9^\circ)$

5. For $3-5i$, find its absolute value and convert it to trigonometric form (round θ to the nearest tenth).

$$3 - 5i$$

→ Compute $r = \sqrt{3^2 + (-5)^2} = \sqrt{9 + 25} = \sqrt{34}$.

→ Use $\tan(\theta) = -5/3$, giving a reference angle of about 59.1° .

→ Since $3-5i$ lies in Quadrant IV, $\theta = 360^\circ - 59.1^\circ = 300.9^\circ$.

→ Write $\sqrt{34}(\cos 300.9^\circ + i \sin 300.9^\circ)$.

Answer: $\sqrt{34}(\cos 300.9^\circ + i \sin 300.9^\circ)$



6. Convert the complex number to trigonometric form.

$$1 + i$$

→ Find $r = \sqrt{1^2 + 1^2} = \sqrt{2}$.

→ Use $\tan(\theta) = 1/1 = 1$, giving $\theta = 45^\circ$.

→ The point is in Quadrant I, so $\theta = 45^\circ$.

→ Write $\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$.

Answer: $\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$

7. Convert the complex number to trigonometric form.

$$-2 + 2\sqrt{3}i$$

→ Find $r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$.

→ Reference angle: $\tan^{-1}(2\sqrt{3}/2) = \tan^{-1}(\sqrt{3}) = 60^\circ$.

→ Since the point is in Quadrant II, $\theta = 180^\circ - 60^\circ = 120^\circ$.

→ Write $4(\cos 120^\circ + i \sin 120^\circ)$.

Answer: $4(\cos 120^\circ + i \sin 120^\circ)$

8. Convert the complex number to trigonometric form.

$$-3 - 3i$$

→ Find $r = \sqrt{(-3)^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$.

→ Reference angle: $\tan^{-1}(3/3) = 45^\circ$.

→ Since the point lies in Quadrant III, $\theta = 180^\circ + 45^\circ = 225^\circ$.

→ Write $3\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$.

Answer: $3\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$

9. Find the absolute value of the complex number.

$$|6 - 8i|$$

→ Apply $|a+bi| = \sqrt{a^2 + b^2}$.

→ Substitute $a = 6$ and $b = -8$ to get $\sqrt{36 + 64}$.

→ Simplify $\sqrt{100} = 10$.

Answer: 10

10. Convert the complex number to trigonometric form (round θ to the nearest tenth).

$$5 - 12i$$

→ Find $r = \sqrt{5^2 + (-12)^2} = \sqrt{169} = 13$.

→ Reference angle: $\tan^{-1}(12/5) \approx 67.4^\circ$.

→ Since the point lies in Quadrant IV, $\theta = 360^\circ - 67.4^\circ = 292.6^\circ$.

→ Write $13(\cos 292.6^\circ + i \sin 292.6^\circ)$.

Answer: $13(\cos 292.6^\circ + i \sin 292.6^\circ)$

